Project Initialization and Planning Phase

Date	30 June 2026
Team ID	******
Project Title	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform loan approval using machine learning, boosting efficiency and accuracy. It tackles system inefficiencies, promising better operations, reduced risks, and happier customers. Key features include a machine learning-based credit model and real-time decision-making.

Project Overview	
Objective	Develop an AI-powered crop recommendation system that predicts the most suitable crop based on soil nutrients and environmental conditions using machine learning.
Scope	The project collects soil and climate parameters, processes them using a trained Random Forest model, and provides crop recommendations through a Flask backend and React frontend.
Problem Statement	
Description	Farmers often struggle to select the right crop due to changing environmental conditions and limited access to data-driven decision support.
Impact	Accurate crop recommendations improve productivity, reduce farming risks, optimize resource utilization, and support sustainable agriculture.
Proposed Solution	
Approach	Use a Random Forest machine learning model trained on agricultural data. The Flask backend processes user inputs and the React frontend displays crop recommendations.
Key Features	• AI-based crop recommendation • Real-time prediction

	• User-friendly web interface • Accurate Random Forest model

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	Processor	Intel Core i3 or above
Memory	RAM	8 GB
Storage	Disk Space	10 GB Free Space
Software		
Frameworks	Backend	Flask
Libraries	Frontend	React.js
Development Environment	ML Libraries	Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn
Data		
Dataset	Agricultural Crop Dataset	CSV Dataset (~2200 records)